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PHNOM PENH INTERNATIONAL UNIVERSITY

Big data Term Project – Project Charter

GENERAL PROJECT INFORMATION

PROJECT NAME

UrbanCart reports

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COURSE / CLASS

Big data

UrbanCart

Ecommerce Data Engineering & Analytics Report

Prepared from ecommerce.db, the legacy customer export, and the 2024 product catalog. All figures in this report are computed directly from the underlying data — none are estimated or carried over from prior specification documents.

Scope note: the UrbanCart Master Specification document supplied alongside the data contains unverifiable infrastructure claims and pre-stated results (e.g. specific query-time reductions, named cloud services, dollar-value impact figures) that could not be reconciled with the actual dataset. This report instead follows the README data dictionary and reports only what was independently computed from the source data.

Executive Summary

The dataset covers 2,500 customers, 9,000 orders, 20,362 order line items, 300 products, 4,000 reviews, and 12,000 web sessions, spanning 34 months of completed-order history.

- Of 9,000 orders, 5,101 (56.7%) are completed; 1,328 cancelled and 1,327 returned.
- Data quality issues found and corrected: 48 duplicate customer records, 186 duplicate order-item rows, 583 negative-quantity line items, and 41 out-of-range review ratings.
- Average customer gross margin rate is 48.8%; mean predicted CLTV is \$1,916.85 (median \$990.27).
- "Champions" is the highest-value RFM segment (avg CLTV \$3,257.53); "Hibernating/Lost" customers still represent 535 people worth reactivating.
- Regression and price-elasticity models show weak/near-zero explanatory power outside of purchase frequency — consistent with the order, pricing, and quantity fields behaving as statistically independent rather than reflecting real consumer price sensitivity.
- A Holt's linear-trend forecast projects monthly revenue rising from a historical average of \$212,712.43 to roughly \$355,853.94 within 6 months, driven mainly by the fitted upward trend rather than seasonality.

Methodology & Data Sources

Three source inputs were used:

- ecommerce.db — SQLite database with customers, products, orders, order_items, reviews, and web_sessions tables (source of truth).
- legacy_customers_export.csv — a legacy customer file with inconsistent name casing, four different date formats, and duplicate/blank rows, reconciled against the customers table.
- product_catalog_2024.csv — a supplier-side catalog with different column names and prices than the internal products table, reconciled by SKU.

All analysis code is organized as a modular Python pipeline (src/) with SQL executed via sqlite3/pandas, statistics implemented with numpy (closed-form OLS, Holt's exponential smoothing, cosine similarity) rather than black-box libraries, and fuzzy matching via rapidfuzz. Every figure in this report is pulled directly from the pipeline's JSON/CSV outputs — see the Appendix for the file manifest.

1. Data Quality Audit

1.1 Core Database Tables

Table	Rows	Key issues found
customers	2500	63 missing city, 139 missing age, 115 missing gender
products	300	3 price outliers (IQR method), 0 cost > price rows
orders	9000	Status split: completed 5101, cancelled 1328, returned 1327, pending 1244
order_items	20362	186 exact duplicate rows, 583 negative-quantity rows (returns)
reviews	4000	838 missing review text, 41 ratings outside 1-5 range

web_sessions	12000	No missing values or duplicates found
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Referential integrity: zero orphaned rows across all foreign-key relationships (orders→customers, order_items→orders/products, reviews→customers/products).

1.2 Legacy Customer Export (raw)

- 1,427 raw rows across 5 columns with inconsistent naming (Customer Name, EMAIL_ADDR, Signup_Dt, Home City, Marketing Segment).
- 57 rows missing an email address; 1 fully blank row; 1 row identified as a test account.
- Signup dates appeared in four different formats within the same column (ISO, 'Month DD, YYYY', MM/DD/YYYY, DD-Mon-YYYY), all successfully parsed.

1.3 Product Catalog 2024 (raw)

- 267 rows using supplier-side column names (SKU, item_name, dept, list_price_usd, supplier_cost, in_stock_units).
- 12 SKUs (IDs 9000–9011) do not exist in the internal products table — likely new/discontinued supplier items never synced to the catalog DB.

2. Cleaning & Reconciliation

2.1 Legacy Customers

Step	Result
Raw rows	1427
Blank rows dropped	1
Test/junk accounts dropped	1
Exact duplicate emails dropped	48
Near-duplicate name pairs flagged (no email to disambiguate)	2
Rows after cleaning	1377
Matched to an existing customers.db record (by email)	1321
Legacy-only / unmatched customers	56

Cleaning steps applied: whitespace trimming, name casing normalized to Title Case, emails lower-cased, dates parsed into ISO 8601 regardless of source format, and city names title-cased. Output: cleaned_data/legacy_customers_cleaned.csv.

2.2 Product Catalog

- 255 of 267 catalog SKUs matched an internal product_id; 12 exist only in the supplier catalog.
- Among overlapping SKUs, 253 (99%) had a list price that differs from the internal unit_price, by an average of \$9.44. This is a systematic, near-universal discrepancy — worth a pricing-source reconciliation with the merchandising team rather than treating as noise.

Output: cleaned_data/product_catalog_cleaned.csv and cleaned_data/catalog_vs_db_price_comparison.csv (full line-by-line diff).

3. Cohort Retention & Market Basket Analysis (SQL)

3.1 Monthly Cohort Retention

Retention rate = distinct customers from a signup cohort placing a completed order N months later, divided by cohort size. Averaged across all cohorts with at least 12 months of history:

Months since signup	Avg. retention
Month 0	7.0%
Month 1	11.0%
Month 2	12.8%
Month 3	12.0%
Month 4	9.8%
Month 5	10.4%
Month 6	9.6%
Month 7	10.5%
Month 8	8.8%
Month 9	8.9%
Month 10	9.7%
Month 11	8.8%

The curve stays flat in the 8–13% band rather than showing the typical steep early drop-off, suggesting order timing in this dataset isn't strongly coupled to signup date — full matrix in `outputs/cohort_retention_matrix.csv`.

3.2 Market Basket Analysis

Computed via self-join of `order_items` on shared `order_id`, at both SKU-pair and category-pair granularity (support, confidence, lift).

Top 5 SKU pairs by lift

Product A	Product B	Support	Confidence (A→B)	Lift
When Supplement	Machine Media	0.00045	0.07407	10.475
Black Camera	Coach Furniture	0.00056	0.07813	10.0872
Movement Headphone	Ready Cycling	0.00045	0.06154	9.7901
Court Camera	Owner Non-Fiction	0.00045	0.06897	9.4525
Level Decor	Property Non-Fiction	0.00045	0.07018	9.0608

Top 5 category pairs by lift

Category A	Category B	Support	Lift
Apparel	Sports & Outdoors	0.0855	0.8841
Beauty & Health	Home & Kitchen	0.0868	0.8757
Books & Media	Sports & Outdoors	0.0864	0.8687
Beauty & Health	Sports & Outdoors	0.0827	0.8663
Electronics	Sports & Outdoors	0.0842	0.8588

All category-pair lift values are ≤ 1 , meaning no category combination is purchased together more than chance would predict — cross-sell bundling by category isn't supported by this data. The SKU-level lift values above are driven by very small order counts (3–5 shared orders) and should be treated as exploratory, not a merchandising recommendation, until validated on more data.

4. RFM Segmentation & Customer Lifetime Value

4.1 RFM Segments

Segment	Customers	Avg. Recency (days)	Avg. Frequency	Avg. Monetary (\$)
Loyal Customers	707	179.07	2.52	3,327.93
Hibernating/Lost	535	988.07	0.45	280.6
Champions	441	78.68	3.77	6,934.79
Needs Attention	416	496.25	1.32	2,878.77
At Risk (High Value)	210	561.61	2.86	4,192.84
New/Promising	191	99.77	1.07	1,350.83

Segmentation logic: each customer is scored 1–5 on Recency, Frequency, and Monetary quintiles, then bucketed by rule (e.g. Champions = top quintile on all three; Hibernating/Lost = bottom quintile on all three). Full customer-level scores in `cleaned_data/rfm_segments.csv`.

4.2 Customer Lifetime Value

- $CLTV = \text{avg. order value} \times \text{purchase frequency/year} \times \text{tenure (years)} \times \text{gross margin rate (48.8\%)}$.
- Mean predicted CLTV: \$1,916.85 | Median: \$990.27 | Top decile threshold: \$3,535.4.

Segment	Avg. predicted CLTV (\$)
Champions	3,257.53
At Risk (High Value)	2,734.09
Needs Attention	1,953.3
Loyal Customers	1,698.55
New/Promising	575.71
Hibernating/Lost	391.64

Output: `cleaned_data/cltv_predictions.csv` (per-customer).

5. Regression, Price Elasticity & Revenue Simulation

5.1 Regression: drivers of customer monetary value

Closed-form OLS (numpy normal equations) predicting a customer's total completed-order spend from age, purchase frequency, and recency. $n = 2060$, $R^2 = 0.0812$, adjusted $R^2 = 0.0798$.

Predictor	Coefficient (β)	t-statistic
Intercept	-37.5091	-0.061
age	-8.2099	-0.893
frequency	1707.4544	12.668
recency_days	0.2501	0.395

Only purchase frequency is a statistically meaningful predictor ($t = 12.7$); age and recency are not significant. The low R^2 (0.08) indicates spend is driven mostly by factors outside this feature set (or by "The dataset exhibits highly regular and weakly coupled statistical patterns, including evenly distributed product categories and payment methods, which are consistent with a controlled or synthetic dataset. However, the dataset files themselves do not provide sufficient evidence to conclusively establish their generation process).

5.2 Price Elasticity of Demand

Log-log OLS of quantity on effective (post-discount) unit price, by product category:

Category	Elasticity	t-stat	R^2	n
Apparel	0.0115	1.4	0.0011	1862
Beauty & Health	-0.005	-0.646	0.0002	1866
Books & Media	0.0051	0.846	0.0004	1903
Electronics	-0.0096	-1.121	0.0007	1832
Home & Kitchen	-0.0121	-1.32	0.0009	1914
Sports & Outdoors	-0.0003	-0.032	0	1815

Overall elasticity: -0.0002 ($t = -0.062$, $R^2 = 0$, $n = 11192$). All elasticities are statistically indistinguishable from zero — quantity purchased shows no measurable relationship to price in this dataset, so elasticity-based pricing recommendations aren't supportable from this data as-is.

5.3 Monte Carlo Revenue Simulation

10,000 simulations of the next 12 months, drawing monthly revenue from a Normal($\mu = \$212,712.43$, $\sigma = \$94,177.14$) distribution fit to historical completed-order revenue.

Metric	Value
Simulated mean 12-mo revenue	\$2,553,111.74
P10 (downside case)	\$2,132,708.39
P50 (median)	\$2,547,399.71
P90 (upside case)	\$2,977,445.29

6. Recommendation Engine

Item-based collaborative filtering: a binary customer $\times$ product purchase matrix (2180 customers $\times$ 300 products, 98.32% sparse) with cosine similarity computed between product columns using numpy (no external ML library). Customer recommendations are generated by scoring each product as (purchase vector) $\cdot$ (item-similarity matrix), excluding already-purchased items.

Example: products similar to "Black Camera"

Similar product	Cosine similarity
Rich Furniture	0.1031
Important Cycling	0.0937
High Camping	0.0937

Full similar-items lookup for all products: cleaned_data/product_similar_items.csv. Per-customer recommendations are generated the same way at query time from the same similarity matrix.

7. Revenue Forecasting

34 months of historical completed-order revenue, averaging \$212,712.43/month. Two models: Holt's linear exponential smoothing ($\alpha=0.4$, $\beta=0.2$, implemented from scratch with numpy) and an OLS linear-trend baseline (slope \$7,752.98/month).

Model	In-sample MAE	In-sample RMSE
Holt's linear	\$56,990.39	\$72,524.81
Linear trend (OLS)	\$42,291.66	—

6-month forward forecast

Month	Holt forecast	Linear trend forecast
2024-12	\$334,411.01	\$348,389.54
2025-01	\$338,699.6	\$356,142.52
2025-02	\$342,988.18	\$363,895.5
2025-03	\$347,276.77	\$371,648.48
2025-04	\$351,565.35	\$379,401.45
2025-05	\$355,853.94	\$387,154.43

Both models agree on direction (revenue rising) but diverge in magnitude by month 6 (~\$334K vs ~\$387K) — the gap is the trend-extrapolation risk inherent to only 34 months of history with no clear seasonal pattern. Treat these as a range, not a point estimate.

8. Conclusions & Recommendations

- Prioritize reactivating the 535 'Hibernating/Lost' customers and the 210 'At Risk (High Value)' customers (\$2,734 avg CLTV) before they fully churn — this is the largest addressable value pool identified.
- Resolve the systematic catalog-vs-internal price mismatch (253 of 255 overlapping SKUs, avg \$9.44 diff) with the merchandising/supplier team; this looks like a stale sync issue, not natural price variation.
- Treat the market-basket and price-elasticity results as inconclusive given near-chance lift values and statistically insignificant elasticity — re-run once more transaction history accumulates, or investigate whether order data reflects real customer behavior vs. system-generated test data.
- Use the Monte Carlo P10–P90 range (\$2,132,708.39–\$2,977,445.29) rather than a single forecast number for 12-month revenue planning.

Appendix: File Manifest

Code (src/)

- audit.py
- clean.py
- sql_analysis.py
- rfm_cltv.py
- regression_elasticity.py
- recommend.py
- forecast.py
- main.py

Cleaned data (cleaned_data/)

- legacy_customers_cleaned.csv
- product_catalog_cleaned.csv
- catalog_vs_db_price_comparison.csv
- rfm_segments.csv
- cltv_predictions.csv
- product_similar_items.csv

Analysis outputs (outputs/)

- audit_report.json
- cleaning_report.json
- sql_analysis_summary.json
- cohort_retention_matrix.csv
- market_basket_top_pairs.csv
- category_affinity.csv
- rfm_cltv_summary.json
- rfm_segment_summary.csv
- regression_elasticity_montecarlo.json
- historical_monthly_revenue.csv
- recommendation_engine_summary.json
- forecast_summary.json
- revenue_forecast_next_6mo.csv